

Book Catalogue API - API documentation

Base URL (local default): `http://127.0.0.1:8000`

API prefix: `/api/v1`

Media type: `application/json`

Authentication: Optional. If the server is configured with environment variable `API_KEY`, then `POST`, `PATCH`, and `DELETE` must include header `X-API-Key: <same value>`. If `API_KEY` is not set (default for local development), writes do not require a key. `GET` endpoints are always public.

Version: 1.2.0 (see `GET /` and `app/config.py`).

Conventions

- **Success:** JSON body with `Content-Type: application/json`.
 - **Validation errors:** HTTP 422 with FastAPI default error shape (detail array).
 - **Not found:** HTTP 404, body `{"detail": "Book not found"}`.
 - **Conflict (duplicate ISBN):** HTTP 409, body `{"detail": "ISBN already exists for another book."}`.
 - **Unauthorized (API key):** HTTP 401, body `{"detail": "Missing or invalid X-API-Key header."}` when `API_KEY` is set but `X-API-Key` is missing or wrong (writes only).
 - **Service unavailable:** HTTP 503 from `GET /health` if the database cannot be reached, body `{"detail": "Database unavailable"}`.
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1. Service metadata

`GET /`

Response 200

```
{
  "service": "Book Catalogue API",
  "version": "1.2.0",
  "repository": "https://github.com/RuncongZhou/xjco3011-coursework-api",
  "docs": "/docs",
  "openapi": "/openapi.json",
  "api_base": "/api/v1",
  "write_protection": false
}
```

(`write_protection` is true when `API_KEY` is configured.)

2. Health

`GET /health`

Response 200

```
{ "status": "ok", "database": "connected" }
```

Response 503 (database unreachable)

```
{ "detail": "Database unavailable" }
```

3. Create book

POST /api/v1/books

Headers (when API_KEY is set): X-API-Key: <your API key>

Request body

Field	Type	Required	Notes
title	string	yes	1-500 chars
author	string	yes	1-300 chars
isbn	string	no	max 20 chars; unique if set
publication_year	integer	no	1000-2100
genre	string	no	max 120 chars
pages	integer	no	>= 0
rating	number	no	0-10
notes	string	no	optional free text

Example request

POST /api/v1/books HTTP/1.1

Content-Type: application/json

```
{
  "title": "The Pragmatic Programmer",
  "author": "David Thomas",
  "isbn": "978-0135957059",
  "publication_year": 2019,
  "genre": "Technology",
  "pages": 352,
  "rating": 9.2
}
```

Response 201

```
{
  "title": "The Pragmatic Programmer",
  "author": "David Thomas",
  "isbn": "978-0135957059",
  "publication_year": 2019,
  "genre": "Technology",
  "pages": 352,
  "rating": 9.2,
  "notes": null,
  "id": 1,
  "created_at": "2026-04-14T12:00:00Z",
  "updated_at": "2026-04-14T12:00:00Z"
}
```

4. List books

GET /api/v1/books

Query parameters

Name	Type	Default	Description
skip	int	0	Offset
limit	int	20	Page size (1-100)
genre	string	(none)	Exact match on genre
author	string	(none)	Case-insensitive substring match on author
q	string	(none)	Search substring in title, author, or ISBN (max 200 chars)
sort	string	created_at	One of: created_at , title , rating , publication_year
order	string	desc	asc or desc

Response 200: paginated wrapper:

```
{
  "items": [ { "id": 1, "title": "...", "...": "same shape as create response" } ],
  "total": 42,
  "skip": 0,
  "limit": 20
}
```

Each element of `items` matches the book object returned by `create` (section 3) and `get-one` (section 6).

5. Statistics

GET `/api/v1/books/stats/summary`

Registered as a static path before `/book_id`, so `stats` is not interpreted as a numeric id.

Response 200

```
{
  "total_books": 3,
  "average_rating": 9.1,
  "genres": {
    "Technology": 3
  }
}
```

(`average_rating` may be `null` if no ratings are stored.)

6. Get one book

GET `/api/v1/books/{book_id}`

Response 200: single book object.

Response 404: `{"detail": "Book not found"}`.

7. Update book

PATCH `/api/v1/books/{book_id}`

Headers (when API_KEY is set): X-API-Key: <your API key>

Send only fields to change (all optional). Field constraints match `create` (section 3) where applicable.

Example

```
{ "rating": 9.5, "notes": "Recommended" }
```

Response 200: updated book.

Response 404: not found.

Response 409: ISBN conflict.

8. Delete book

DELETE /api/v1/books/{book_id}

Headers (when API_KEY is set): X-API-Key: <your API key>

Response 204: no body.

Response 404: not found.

Interactive documentation

With the server running, open **Swagger UI** at /docs to try requests from the browser. **ReDoc** is at /redoc. Machine-readable schema: /openapi.json (see also docs/openapi.json from scripts/export_openapi.py).

Export for submission

- **PDF:** print /docs to PDF, export this Markdown from your editor, or run `pandoc docs/API_DOCUMENTATION.md -o docs/API_DOCUMENTATION.pdf` if Pandoc is installed. Commit docs/API_DOCUMENTATION.pdf if your brief requires a PDF in the repo.
- **OpenAPI JSON:** `python scripts/export_openapi.py` writes docs/openapi.json.